Fixing the Dark-Energy Equation of State with Two Dimensionless Constants, φ and π: A Falsifiable, Parameter-Frugal Test Against DESI DR2 and Planck 2018

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Abstract

We present a minimal-parameter framework for the dark sector in which the *shape* of late-time cosmic acceleration is fixed algebraically by two dimensionless constants, the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, rather than by tuned cosmological parameters. The equation of state follows as bare ratios of structural invariants, $w_0 = -T_r/M_v = -0.839950$ and $w_a = -P_{\mathrm{sc}}/I_g = -0.669975$, with no fitted amplitude. The point was fixed in advance and sits at 0.24σ from the DESI DR2 $w_0 w_a$CDM posterior (Pantheon+; 0.2–1.8σ across supernova compilations, $\chi^2_{\mathrm{2D}} = 0.42$). Crucially, the two forward predictions that can *refute* the core framework—this fixed (w_0, w_a) point and the tensor ratio $r = \varphi^{-10}$—stand on their own: neither the Hubble resolution nor the growth sector is invoked to state or to kill them. A Markov-chain analysis combining DESI BAO with the global rate of the screening cascade as prior ($H_0^{\mathrm{glob}} = 67.962 \pm 0.968$ km s^{-1} Mpc^{-1}) yields $H_0 = 67.82 \pm 0.41$ km s^{-1} Mpc^{-1} (0.68σ from Planck, 0.34σ from H_0^{glob}). With the background of both models fixed in advance, DESI DR2 BAO gives $\chi^2 = 11.41$ for SSEE and 31.98 for ΛCDM at its Planck values (13 points); ΛCDM reaches 10.54 only by lowering Ω_m to 0.297. The effective-field-theory description is canonical—$\alpha_T = \alpha_M = \alpha_B = 0$ with $\alpha_K = 15.591336$ and a ghost-free sound speed $c_s^2 = 0.021284$—and the inflationary sector predicts $n_s = 1 - \varphi^{-7} = 0.965558$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.130619 \times 10^{-3}$ that is testable by LiteBIRD and CMB-S4. With its single matter density $\Omega_m = \omega_m/h^2 = 0.308881$, fixed by the algebra, a Boltzmann calculation places the first three acoustic peaks at $\ell = 220, 536, 813$, against $220, 536, 813$ for ΛCDM. We are explicit about the framework's limits: the absolute scale enters as a single measured input (the local expansion rate), and the algebraic origin of the CMB matter density (ω_b, ω_c) remains an open problem. The growth amplitude S_8, previously reported here as a standing tension with weak lensing, is not one: against the 357 raw KiDS-Legacy $\xi_\pm$ points, with a single matter sector, the background fixed by the algebra and A_s fixed to the value the model's own CMB fit gives — *no* free cosmological parameter — it predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured (0.04σ); the amplitude the shear would ask for on its own sits 0.49σ from the CMB one. The earlier test against raw KiDS-1000, with A_s fitted, gave 0.7559 ± 0.0189, statistically indistinguishable from the ΛCDM control on the same raw data (0.7571 ± 0.0194), with $\Delta\chi^2 = 2.69$ for four *fewer* free parameters ($\Delta\mathrm{BIC} = -19.0$ favouring SSEE). The φ-dark-matter extension that was formerly kept outside the core register is retracted. Both core forward predictions bear on *unreleased* data: the (w_0, w_a) point is pre-registered to be excluded at $> 3\sigma$ if DESI DR3 shifts it, and $r = \varphi^{-10} = 8.130619 \times 10^{-3}$ is testable by LiteBIRD—either alone can refute the framework. The result is a falsifiable, parameter-frugal description of dark energy whose core predictions rest on φ and π alone.

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Notation: algebraic constants used in this work

All constants below are *dimensionless* and built solely from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Constants defined in earlier SSEE material but *not used* in this submission (e.g. VITA) are deliberately omitted to keep the register honest.

Each row gives the *symbol* used in the equations, the *name* of the entity in the constant dictionary [Almeida Vallejo, 2026], its *parents* (the entities it is generated from, i.e. its lineage), and the same quantity *reduced to ordinary algebra* in φ and π alone. Symbols are in one-to-one correspondence with entities: no symbol denotes two entities and no entity carries two symbols. The two columns are not redundant— they are the point. I_g and K_v have *different parents* but reduce to the *same* expression $2(\varphi+\pi)$, which is why they share the value 9.519253 while remaining structurally distinct. The value therefore cannot identify which one an equation uses; only the symbol can.

Symbol	Name	Parents	Reduced form in φ, π	Value
φ	—	seed (UV)	$(1+\sqrt{5})/2$	1.618034
π	—	seed (IR)	π	3.141593
Ω	OMEGA	φ, π	$\varphi+\pi$	4.759627
β	BIAL	φ, π	$(\varphi+\pi)/2$	2.379813
KAL	KAL	β, π	$(\varphi+3\pi)/2$	5.521406
AURA	AURA	β, φ	$(3\varphi+\pi)/2$	3.997847
MIRA	MIRA	AURA	$(3\varphi+\pi)/4$	1.998924
P_{sc}	PYROS	Ω, φ	$2\varphi+\pi$	6.377661
I_g	IGNIS	π, PYROS	$2(\varphi+\pi)$	9.519253
K_v	KRYSTOS$_V$	φ, π, Ω	$2(\varphi+\pi)$	9.519253
T_r	TRIAL	AURA	$3(3\varphi+\pi)/2$	11.993542
M_v	ATLAS	φ, π, K_v	$3(\varphi+\pi)$	14.278880

Roles in the equations: Ω is the stability metric; β the base coupling; KAL the structural retention; AURA the screening amplitude; MIRA the mirror ratio entering f_{screen} (Paper 9); P_{sc} and I_g the numerator and denominator of w_a; T_r and M_v those of w_0; and K_v builds M_v but—this is the point of the parent column—*never* appears as a denominator: the denominator of w_a is I_g, generated from its own numerator by $I_g = \pi + P_{\text{sc}}$.
Derived background quantities (all dimensionless):

$$w_0 = -\frac{T_r}{M_v} = -0.839950, \qquad w_a = -\frac{P_{\text{sc}}}{I_g} = -0.669975, \qquad s_m = 1 + w_0 = 0.160050.$$

1 Introduction

The standard cosmological model, ΛCDM, fits the available data with six free parameters [Planck Collaboration, 2020a], but pays for this success with a dark sector whose composition and energy scale remain unexplained. The recent DESI DR2 measurements, which favour a dynamical, evolving dark-energy equation of state over a pure cosmological constant [Abdul-Karim et al., 2025], sharpen the question: if dark energy is dynamical, what fixes its evolution? A theory that could *predict* the equation of state, rather than fit it, would mark genuine progress.

This paper develops one such proposal. The central hypothesis of the Structural Self-Energy Expansion (SSEE) framework is that the *shape* of the dark sector—the equation-of-state ratios, the effective-field-theory function profile, and the matter density ω_m—is fixed algebraically by two dimensionless constants: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. From these, a small set of structural invariants (Table) is built, and the late-time observables follow as bare ratios with no tuned amplitude.

We are deliberate about what this does and does not claim. SSEE is a *minimal-parameter* framework, not a zero-parameter one: the absolute scale enters as a single measured input, the local expansion rate of SH0ES (Postulate D, §7), reducing the count of free CMB inputs from ΛCDM's six to two (A_s and τ; $k = 2$ vs $k = 6$). The dark-sector *shape*, however, carries no free parameter, and this is what makes the framework falsifiable. Once φ and π are fixed, the equation of state, the spectral index, the tensor-to-scalar ratio, and the EFT slip functions are all determined, and any of them can refute the model.

The paper is organised as a single coherent argument rather than a collection of results. Section 2 constructs the equation of state from the algebraic invariants and confronts it with DESI DR2, derives the single matter density $\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2 = 0.308881$ and separates it from the equation-of-state number $1 + w_0 = 0.160050$, and reports the Hubble-constant inference. Section 4 gives the canonical effective-field-theory description, the inflationary predictions, and the confrontation with Planck 2018, including a Boltzmann cross-check with the model's single matter density (ω_m-direct, $\Omega_{m,\rm CMB} = 0.308881$). Section 5 treats the growth of structure, including the correction that removed the S_8 tension reported by earlier versions. Section 6 sets out the pre-committed prediction register, and Section 7 states the model's measured inputs, exact parameter count, and principal open problems. We conclude in Section 8.

2 The algebraic core: from φ, π to the equation of state

2.1 Construction of the equation-of-state parameters

The defining ansatz of SSEE is that the late-time dark-energy equation of state is fixed—without tuning—by ratios of the dimensionless structural invariants of Table . Concretely,

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} = -0.839950, \qquad w_a = -\frac{P_{\rm sc}}{I_g} = -\frac{\Omega+\varphi}{\pi+\Omega+\varphi} = -0.66997. \tag{1}$$

Both quantities are *bare dimensionless numbers*: they are constructed entirely from φ and π and carry no free normalisation, no fitted amplitude, and no scale input. They enter as the two numbers of the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1-a)$. The field of Paper 7 fixes w_0 and the sound speed today but not $w(a)$: it rolls towards $w = -1$, and a single canonical field does not cross it, so the dynamical origin of w_a is open (OP-24, §4.1).

2.2 Agreement with DESI DR2

We confront Eq. (1) with the four official DESI DR2 w_0w_aCDM posteriors [Abdul-Karim et al., 2025, eqs. 25–28]. The tightest combination, DESI + CMB + Pantheon+, gives $w_0^{\rm bf} = -0.838 \pm 0.055$, $w_a^{\rm bf} = -0.62^{+0.22}_{-0.19}$ (correlation $\rho = -0.894$, measured directly from the official DESI DR2 chains, Zenodo record 16644577, burn-in 30%, chain weights). Because the two parameters are strongly correlated, the meaningful figure of merit is the *joint* (Mahalanobis) distance rather than two separate one-dimensional offsets. Writing $\mathbf{d} = (w_0 - w_0^{\rm bf},\, w_a - w_a^{\rm bf})$ and the 2×2 covariance C,

$$\chi^2_{\rm 2D} = \mathbf{d}^\top \mathsf{C}^{-1}\mathbf{d} = 0.42, \qquad p = 0.81, \qquad \text{equivalent } 0.24\sigma. \tag{2}$$

The component offsets are $\Delta w_0 = -0.04\sigma$ (essentially exact) and $\Delta w_a = -0.26\sigma$. Against the other official combinations (moments and correlations likewise measured from the official chains: $\rho = -0.905$ to -0.978) the joint distance is 1.46σ (DESY5), 1.76σ (Union3) and 1.68σ (no-SN DESI+CMB): the algebraic point is consistent with all four, with the tension range 0.2–1.8σ set by the supernova compilation. The correlation is the one measured from the chains, not an assumed value. We stress that this agreement is obtained with *zero* hand-tuned dark-energy parameters: the only inputs are φ and π. The fixed ΛCDM point $(-1, 0)$, by contrast, sits at 2.6–3.9σ from every official compilation (Table 1): SSEE is closer to the data than ΛCDM for all four supernova compilations—a like-for-like comparison of two zero-parameter predictions against the same measurement, neither of which tunes (w_0, w_a).

Table 1: Mahalanobis distance of the fixed SSEE point $(-0.840, -0.670)$ and the fixed ΛCDM point $(-1, 0)$ from each official DESI DR2 w_0w_aCDM best-fit (2×2 covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than ΛCDM for all four supernova compilations.

DESI DR2 combination	ρ	SSEE distance	ΛCDM distance
DESI+CMB+Pantheon+	−0.894	$\mathbf{0.24\sigma}$	2.58σ
DESI+CMB+DES Y5	−0.905	$\mathbf{1.46\sigma}$	3.93σ
DESI+CMB+Union3	−0.933	$\mathbf{1.76\sigma}$	3.38σ
DESI+CMB (no SN)	−0.978	$\mathbf{1.68\sigma}$	2.62σ

Honest caveat: this is agreement with BAO+SN, not with CMB-only. The 0.24σ figure is specific to the DESI DR2 dynamical-dark-energy posterior, which is dominated by low-redshift data ($z \simeq 0.3$–2.3), where dark energy actually drives the expansion, and the quoted range 0.2–1.8σ reflects the known spread among supernova compilations. Against a *Planck CMB-only* w_0w_aCDM analysis, which constrains the late-time equation of state only through its extrapolation from $z \simeq 1100$, the same (w_0, w_a) is disfavoured. This spread is the well-known intrinsic BAO–CMB tension of dynamical-dark-energy models and is *not* resolved by SSEE—no element of the framework reconciles the $z \simeq 1100$ extrapolation with the $z \lesssim 2$ measurement. We report the DESI agreement as what it is: a clean, parameter-free match to the dataset that directly probes the dark-energy-dominated epoch.

2.3 A look-elsewhere test: how special are these two ratios?

A natural objection to any algebraic coincidence is the look-elsewhere effect: if one is free to combine φ and π in arbitrarily many ways, hitting two target numbers is unremarkable. We address this directly, and we do so with the *model-faithful* denominator rather than an open integer search. SSEE does not draw w_0 and w_a from an unbounded space of expressions; it draws them from a *closed dictionary* of 55 named structural invariants—the *complete* Genesis set, not a hand-picked subset—specified as a fixed set rather than expanded to accommodate the data.[1] The honest question is therefore: among *all* ratios A/B of named invariants in this fixed dictionary, how many land on $|w_0| = 0.839950$ and on $|w_a| = 0.669975$?

Enumerating every ordered pair over the complete dictionary yields 490 distinct ratios in the interval $(0, 5]$. We then count how many fall within a tolerance of each target. Crucially, the SSEE assignments are not near-misses but *exact* algebraic identities—$|w_0| = \mathrm{AURA}/\Omega = T_r/M_v$ and $|w_a| = P_{\mathrm{sc}}/\mathrm{IGNIS} = \mathrm{PYROS}/(\pi + \mathrm{PYROS})$, both to machine precision ($|d| = 0$)—so the physically relevant tolerance is that of the exact identity (± 0.0005, $|d| = 0$), not an arbitrary band. At that precision each target is unique:

$$|w_0| = 0.839950 : \ 1 \text{ of } 490, \qquad |w_a| = 0.669975 : \ 1 \text{ of } 490 \quad (\pm 0.0005). \tag{3}$$

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[1] The dictionary is the full Genesis constant set [Almeida Vallejo, 2026]; the look-elsewhere count below is performed over *every* named invariant, with none enlarged, pruned, or selected to improve the match. Counting over the complete set rather than a subset is the conservative choice: it can only inflate the denominator of accidental hits, never shrink it. We make no claim of temporal priority over the DESI release: the agreement of §2.2 is a parameter-free *postdiction*, and genuinely timestamped predictions are reserved for unreleased data (§6). Many invariants share a value by construction—the 55 names collapse to 25 distinct numbers (e.g. IGNIS and K_v, SOLAR = MAR, and the convergence at 3Ω); the count below deduplicates by value, so each distinct number is counted once.

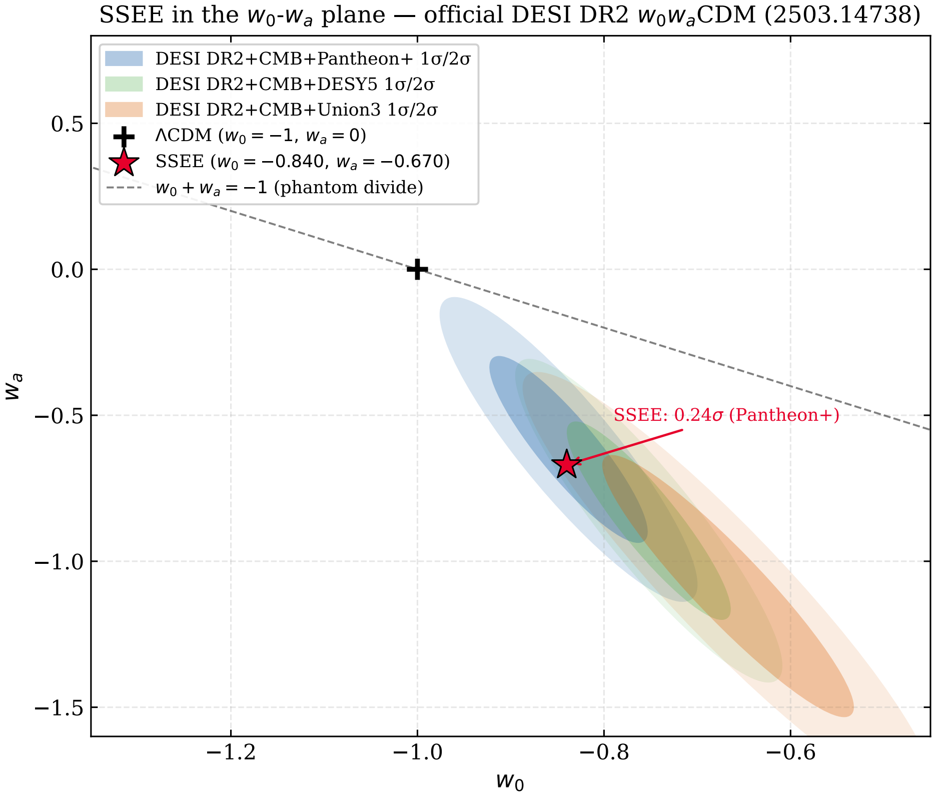

Figure 1: The SSEE prediction in the w_0–w_a plane. The single point $(w_0, w_a) = (-T_r/M_v, -P_{\rm sc}/I_g) = (-0.840, -0.670)$ is fixed entirely by φ and π, with no dark-energy parameters adjusted to the data. The shaded contours are the official DESI DR2 w_0w_aCDM posteriors (Pantheon+, DESY5, Union3). The SSEE point falls inside the Pantheon+ 1σ region; the joint Mahalanobis distance is $\chi^2_{\rm 2D} = 0.42$, an equivalent 0.24σ agreement (Eq. 2), and within 1.5–1.8σ of the other compilations.

Table 2 reports the full sensitivity to the tolerance, and we state it honestly: the uniqueness is a property of the strict, sub-percent regime that the *exact* match actually occupies; relaxing the band far above the DESI precision predictably densifies accidental hits. Within the model's own vocabulary, at the precision of the measurement, each equation-of-state parameter is *essentially unique*—the opposite of a dense forest of coincidences, and a stronger statement than a generic $(a\varphi + b\pi)/(c\varphi + d\pi)$ scan, which by construction inflates the denominator with expressions the model never uses.

The two ratios are moreover not independent draws. Each is an intra-lineage ratio of the closed dictionary: $|w_0| = \mathrm{AURA}/\Omega$ is the copy-branch root over the scaffold, and $|w_a| = \mathrm{PYROS}/\mathrm{IGNIS}$ has a denominator generated directly from its own numerator by the no-self-sum law ($\mathrm{IGNIS} = \pi + \mathrm{PYROS}$). The two CPL numbers are therefore not two separately fitted constants: the joint look-elsewhere is governed by a single shared dictionary rather than by two free choices. We report this as evidence of internal economy, not as a claim that the dictionary itself is derived from first principles—that derivation (the dynamical origin of the two lineage laws) remains the open mechanism tracked in §7.

The single near-value, and why it is not a rival. Honesty requires naming the one entry that turns the $|w_a|$ count from 1 to 2 as the band is loosened from ±0.0005 to ±0.001 (Table 2): it is $\mathrm{SOLAR}/\mathrm{NYX} = 0.66933$, which sits $|d| = 6.4 \times 10^{-4}$ from $|w_a| = \mathrm{PYROS}/\mathrm{IGNIS} = 0.669975$. Two independent facts demote it from rival to shadow. *First, construction:* the canonical ratio PYROS/IGNIS is intra-lineage—the denominator is generated directly from the numerator,

Table 2: Look-elsewhere sensitivity to the matching tolerance, computed over the *complete* 55-constant Genesis dictionary (490 distinct ratios in $(0, 5]$; constants sharing a value collapse to 25 distinct numbers). The DESI assignments are exact identities ($|d| = 0$); the relevant band is ± 0.0005 (the exact-identity precision), where each parameter is unique. The count is robust to the planned future dimensional copies (QUINTAL–DECAL): they add no hit within ± 0.0005 of either target.

Tolerance	hits on $\lvert w_0 \rvert = 0.839950$	hits on $\lvert w_a \rvert = 0.669975$
± 0.0005	1	1
± 0.001	1	2
± 0.002	1	4
± 0.005	2	5
± 0.01	4	9

IGNIS $= \pi +$ PYROS, so numerator and denominator lie on the same π-branch chain—whereas SOLAR and NYX $= (\varphi + 7\pi)/2$ (the retention-branch floor, charge $q = -3$) share no generative relation; their ratio is a numerical coincidence across lineages, not a structural identity. The selection is therefore made by lineage, not by tolerance. *Second, transcendence:* the proximity is the shadow of the numerical near-coincidence $2\varphi \approx \pi$ ($2\varphi = 3.236$, $\pi = 3.142$), the exact residual scaling as $(2\varphi - \pi)(\varphi - \pi) \neq 0$. Because π is transcendental, $2\varphi - \pi$ is irrational and non-zero for all time, so this near-value is *forbidden* from ever collapsing onto an exact identity—by the very property of π that grounds the no-self-sum law of Appendix A. The law that generates NYX is thus the same law that guarantees it can never become a genuine second identity: the near-miss is structural, not accidental, and disclosing it strengthens rather than weakens the count.

Scope: this test is for the core, not the extension. The look-elsewhere statement above applies to the dark-energy core (w_0, w_a), where the dictionary grammar is fixed by construction rules (no-self-sum plus copy-law) that precede and are independent of the DESI targets. No φ-DM mass multiplier is left to test: that construction is retracted. Its warrant had been *falsifiability* — a pre-registered free-streaming scale that DESI Y3 and Euclid would confirm or exclude. That warrant was insufficient, and the episode is instructive: falsifiability is the minimum requirement for a hypothesis to be scientific, not evidence that the entity it concerns exists.

2.4 The cosmic matter fraction: one matter density

The equation of state (1) fixes s_m (written Ω_m^{dyn} in earlier versions; that name is withdrawn),

$$s_m \;=\; 1 + w_0 \;=\; 0.160050, \tag{4}$$

a number from the equation of state, not a density. It is much lower than the matter density inferred from the acoustic peaks, and it is not meant to match it. Under the ω_m-direct reframe (retiring the former matter-bridge open problem, OP-8; §7) the CMB matter fraction is the standard physical observable, built piece-by-piece from SSEE algebra with *no* bridging factor,

$$\Omega_{\mathrm{m}}^{\mathrm{cosm}} \;=\; \frac{\omega_m}{h^2} \;=\; \frac{\omega_b + \omega_c + \omega_\nu}{h^2} \;=\; \frac{0.14267}{(67.96214/100)^2} \;=\; 0.308881, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{5}$$

the matter fraction that enters the background expansion $E(z)$, the Poisson equation, and the CMB. The equation-of-state number $s_m = 0.160050$ (tested by DESI through w_0) is not a second matter density at all: it is $1 + w_0$, a quantity from the equation of state. There is one matter density, $\Omega_{\mathrm{m}}^{\mathrm{cosm}}$; the "two-$\Omega_{\mathrm{m}}$" accounting of earlier versions is withdrawn.

No bridging factor: the CMB density is built from algebra. Each piece of ω_m is an SSEE algebraic value with no fit: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ and the cold-matter density $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514$ (a forward identity, already in Paper 1; 0.40σ from Planck). The neutrino term completes the total, $\omega_\nu = \Sigma m_\nu^{\text{active}}/C_\nu = 0.000735$, using the same active-neutrino sum $\Sigma m_\nu^{\text{active}} = 0.06849$ eV and the same relic closure constant $C_\nu = 93.14$ eV (PDG, $N_{\text{eff}} = 3.046$),[2] so that

$$\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267, \qquad \Omega_{\mathrm{m}}^{\text{cosm}} = \frac{\omega_m}{h^2} = 0.308881 \quad (h = H_0^{\text{glob}}/100). \tag{6}$$

The neutrino piece is not optional: dropping it returns 0.30729, a 0.5% shift, so a reader who reproduces the number must include all three terms (a three-second check ships in `ssee_core.py`). The earlier presentation tied $\Omega_{\mathrm{m}}^{\text{cosm}}$ to s_m through a factor-of-two bridge MIRA = AURA/2 = 1.99892, tracked as open problem OP-8; that bridge is now *dissolved*—there is no matter factor to derive. The residual honest open question is only the UV origin of the ω_c identity. MIRA survives elsewhere, in the late-time Hubble screening fraction $f_{\text{scr}} = s_K/(3\,\mathrm{MIRA})$ (Paper 9), where it is the "Mirror Ratio".

One matter density. The equation-of-state number $1 + w_0 = 0.160050$ and the matter density $\Omega_{\mathrm{m}}^{\text{cosm}} = \omega_m/h^2 = 0.308881$ are two independent algebraic predictions, not two free numbers and not two matter densities: $1 + w_0$ never enters a density slot. Earlier versions of this paper put it there in a Boltzmann run and reported that the acoustic peaks then degrade; that comparison tested a density the model does not contain, and it is withdrawn.

Figure 2 makes the same point on the drag-epoch sound horizon itself: the physical CMB drag scale set by the full algebraic ω_m is 147.17 Mpc at $\Omega_{\mathrm{m}}^{\text{cosm}} = 0.308881$ (ω_m-direct), consistent with the Planck-inferred 147.09 ± 0.26 Mpc (0.32σ), and matching ΛCDM at the Planck 2018 parameters (147.10 Mpc, same code). The equation-of-state number $1 + w_0 = 0.160050$ does not enter the geometry.

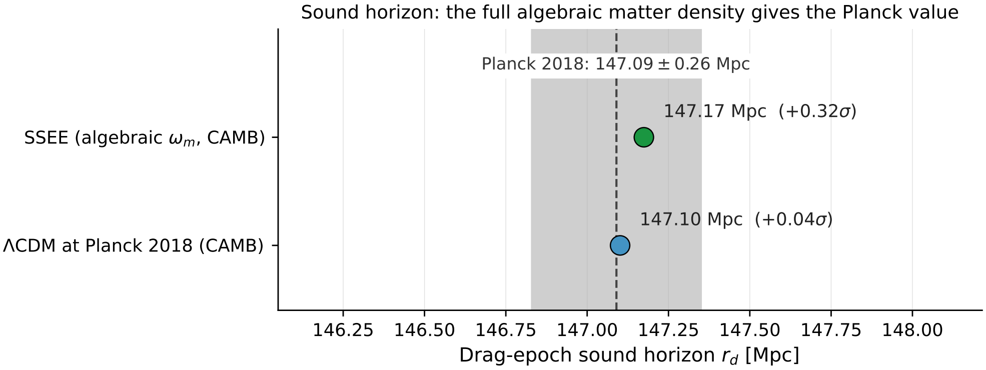

Figure 2: The physical sound horizon. Set by the full algebraic ω_m (green, 147.17 Mpc at $\Omega_{\mathrm{m}}^{\text{cosm}} = 0.308881$, ω_m-direct), in agreement with Planck 2018 (147.09 ± 0.26 Mpc, grey band) and with ΛCDM at the Planck 2018 parameters (blue, 147.10 Mpc, CAMB). This is the visual counterpart of the Boltzmann test above: the full matter density that restores the acoustic peaks also restores the BAO standard ruler.

3 Bayesian validation: MCMC against DESI DR2

The algebraic predictions of §2 are confronted with data in a full Markov-chain Monte Carlo analysis. The dark-energy sector contributes no sampled parameters—w_0, w_a and the matter density

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[2] $\Sigma m_\nu^{\text{active}}$ is not a fitted input: it follows from the algebraic chain $\Sigma m_\nu^{\text{active}} = R_2\, \omega_b\, C_\nu/(\tau_{\Pi} H_0)$ with $R_2 = \Omega/(\mathrm{KAL}_0 T_r)$ (Paper 4). No independent neutrino parameter is introduced.

$\Omega_{\rm m}^{\rm cosm}$ are fixed by Eqs. (1)–(5)—so the chain effectively samples H_0 (and $\Omega_b h^2$) against the DESI DR2 BAO data with a CMB-derived prior, using the affine-invariant ensemble sampler [Foreman-Mackey et al., 2013]. We ran 100 walkers × 25,000 steps (seed 42).

H_0 as one quantity in two stages. We first anchor the early-time geometry at the output of the screening cascade

$$\begin{aligned} H_0^{\rm glob} &= H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968\ {\rm km\,s^{-1}\,Mpc^{-1}}, \\ \frac{H_0^{\rm glob}}{\rm km\,s^{-1}\,Mpc^{-1}} &\simeq 3(\varphi + \pi)^2, \end{aligned} \tag{7}$$

which (with ω_b, ω_c fixed by SSEE algebra) lies 0.90σ from the H_0 that minimises the Planck `plik_lite` likelihood for the SSEE background (67.87 ± 0.10, profiled over A_s, τ); at this anchor the sound horizon and acoustic scale are both well within tolerance, $r_d = 147.17$ Mpc (0.32σ) and $\theta_* = 0.59667^\circ$ (1.00σ). Letting the MCMC then correct H_0 against the low-redshift BAO (corrected DESI DR2 vector, block-diagonal covariance with the official $r_{M,H}$ correlations) yields the posterior

$$H_0^{\rm post} = 67.82 \pm 0.41\ {\rm km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_b h^2 = 0.02198 \pm 0.00048, \tag{8}$$

with $r_d = 147.17$ Mpc (0.32σ, ω_m-set, H_0-invariant) at the posterior value. These are two stages of the same quantity—a CMB anchor and a BAO-corrected posterior—and with the total matter density in the background geometry they *coincide*. (The earlier MIRA-anchored 67.037/66.533 stages, the 67.159 posterior from the mislabelled DR1 vector, and the 66.412 obtained with the cold sector 0.160 in the geometry, are all superseded.)

Status of the algebraic anchor. We are explicit about how a dimensionful value is obtained from Eq. (7), and about which observable is the input. On its own, $3(\varphi + \pi)^2 = 67.962$ is a pure number with no units; deriving the absolute scale of H_0 from first principles is an open problem for every cosmological model, not only ours, and we make no claim to have solved it. We therefore do not assert the pure number as physical and then show that it explains the data. Instead we start from the measured SH0ES *local* rate, $H_0^{\rm SH0ES} = 73.04 \pm 1.04\ {\rm km\,s^{-1}\,Mpc^{-1}}$ [Riess et al., 2022], and, under the screening postulate of Paper 9 (the correction acts only on locally measured rates), de-screen it with the algebraically-derived (not fit) fraction $f_{\rm scr}^{\rm IR} = s_K^{\rm IR}/(3\,{\rm MIRA}) = (\pi - \varphi)/\Omega_{\rm m}^2 = 0.067253$ (Paper 9): $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm IR}) = 68.13\ {\rm km\,s^{-1}\,Mpc^{-1}}$, within 0.17σ of the pure number 67.962—the model-independent floor, using only the IR screening term and no free parameter. Paper 10 sharpens this: with the UV-completed fraction $f_{\rm scr}^{\rm UV} = s_K^{\rm full}/(3\,{\rm MIRA}) = 0.069522$ (with the cutoff $M^4 = 5\varphi^8 \rho_{\rm crit}$ of Conjecture C.1; the cascade is consistent with that cutoff but does not measure it), the identical inversion gives $H_0^{\rm SH0ES}(1 - f_{\rm scr}^{\rm UV}) = 73.04 \times (1 - 0.069522) = 67.96214\ {\rm km\,s^{-1}\,Mpc^{-1}}$, reproducing the pure-number target $3(\varphi + \pi)^2 = 67.9621$ to four decimals. We quote the comparison at four decimals and not further because the input is quoted at two: $H_0^{\rm SH0ES} = 73.04$. We do not quote the reverse operation $3(\varphi + \pi)^2/(1 - f_{\rm scr}^{\rm UV})$ as a prediction, because $3(\varphi + \pi)^2$ is a pure number and carries no units, so it cannot serve as the input of a dimensionful cascade. The only directly measured expansion rate is the local one, and Planck's is inferred within ΛCDM rather than measured; SH0ES is therefore the input throughout. The leftover, $4.2 \times 10^{-6}\ {\rm km\,s^{-1}\,Mpc^{-1}}$, lies below the resolution of the datum it is compared against; whether a future, more precisely quoted local measurement drives it down or opens it up is an empirical question, not one we can settle here. We say "to the quoted precision" rather than "exactly" for that reason. We are deliberately precise about what is independent of what. The de-screening is *external*: it consumes only the measured SH0ES rate and a screening fraction derived beforehand, with no SSEE value of H_0 as input—but its two variants are one datum under two prescriptions, not two independent measurements. The CMB/BAO chain, which never touches SH0ES, adopts the anchor as its prior, so its posterior is a *consistency* test rather than a third confirmation: its content is that DESI DR2 does not pull the value away ($\chi^2_{\rm BAO} = 11.41$ over 13 points at the anchor, against $\simeq 726$ for the incorrect background geometry), leaving a residual displacement of 0.34σ. Replacing the anchor prior with a Planck one removes even that dependence: the data then pull H_0 *upward* from 67.36 to 67.87 ± 0.28, towards the anchor rather than away from it. That pattern—one external datum and two independent-prior checks that decline to move the value—is the anchor's physical content, not a claim that $3(\varphi + \pi)^2$ literally equals a dimensionful rate. In Planck units $H_0 \sim 10^{-61}$ and no such coincidence appears; the dimensional origin—the bridge from the megaparsec scale to fundamental units—remains open (§7). The anchor is falsifiable on both fronts: a CMB/BAO posterior displaced from it by more than 2σ, or a SH0ES-derived local rate that moves the unconditional IR floor outside $\sim 1\sigma$, would undermine it.

Anchor and posterior agree. With the total gravitating matter density $\Omega_{\rm m}^{\rm cosm} = 0.308881$ in $E(z)$ and r_d, the BAO-corrected posterior (67.82) and the CMB anchor $H_0^{\rm glob}$ (67.962) agree to 0.34σ: DESI DR2 is *consistent with* the global rate rather than pulling against it. All early-time observables are simultaneously healthy—$r_d = 147.17$ Mpc (0.32σ) and, evaluating the acoustic scale at the posterior, $100\,\theta_* = 1.04099$ (0.34σ). The apparent 2.9σ "BAO–CMB split" and the 13.9σ θ_* figure reported in earlier drafts were the footprint of a single error—placing the equation-of-state number $s_m = 0.160050$ in the background geometry, where the total density belongs. With that corrected, there is no split to report: the late-time BAO and the early-time acoustic geometry point to the same H_0.

Model comparison against the natural competitors. Because the dark-energy sector carries no sampled parameters, the honest comparison is not against ΛCDM alone but against the whole family a referee would reach for. On the identical DESI DR2 vector, the information criteria are: This is the conservative version: it lets SSEE sample H_0 and $\Omega_b h^2$ and charges it $k = 2$ for them, although ω_b is algebraic and H_0 is the output of the cascade. Evaluated at its algebraic point with nothing free ($k = 0$), SSEE is favoured by $\Delta\text{BIC} = -10.90$ over ΛCDM and -9.32 over CPL (Paper 2); freeing H_0 and $\Omega_b h^2$ lowers χ^2 by only 0.46. In the conservative version SSEE is favoured over both ΛCDM ($\Delta\text{BIC} = -5.81$) and CPL (-4.24) by parsimony, with $\Delta\text{DIC} = -5.02$ and a Savage–Dickey factor $\ln B = 2.379$ (substantial on the Jeffreys scale); a leave-one-out cross-validation at $z \geq 1$ ranks SSEE ahead of ΛCDM ($\chi^2_r = 0.26$ versus 0.71; CPL's training fit runs to the edge of the H_0 prior and is not quoted). The preference is driven by fewer parameters at equal fit, not by a superior χ^2: the competitor that *can* bend to the data (CPL, $k = 5$) is penalised for the freedom SSEE does not use.

Table 3: Bayesian model comparison on DESI DR2 (100 walkers × 25,000 steps), with the compressed Planck prior; conservative version with H_0 and $\Omega_b h^2$ sampled in SSEE. $\Delta \equiv$ SSEE−model; negative Δ favours SSEE. With nothing free ($k = 0$) the preference grows (text).

Model	k	$\ln P_{\rm MAP}$	BIC	ΔBIC	ΔAIC
SSEE	2	−6.24	18.02	0.00	0.00
ΛCDM	3	−7.76	23.83	−5.81	−5.04
CPL	5	−4.20	22.26	−4.24	−1.92

Each probe with the background fixed in advance. The comparison above lets ΛCDM fit its background to DESI. Table 4 instead fixes the background of both models before looking at each probe—SSEE at its algebraic point, ΛCDM at the Planck 2018 best fit—and reads the χ^2 of each probe with no cosmological parameter fitted. One probe separates the two models by more than $\Delta\chi^2 = 9$: DESI DR2 BAO, with $\chi^2 = 11.41$ for SSEE and 31.98 for ΛCDM over 13 points ($\Delta\chi^2 = -20.6$). When ΛCDM is allowed to fit DESI, it reaches $\chi^2 = 10.54$ for 11 degrees of freedom by lowering Ω_m from 0.3152 to 0.297 ± 0.009; SSEE reaches 11.41 with nothing fitted. The other probes do not separate the models. The supernova compilations share objects and the two CMB-lensing probes share sky, so the rows are not summed.

Table 4: χ^2 of SSEE and ΛCDM probe by probe, read from the analysis logs. Block A: background fixed in both models in advance (ΛCDM at the Planck 2018 best fit). Block B: ΛCDM fits cosmological parameters that SSEE does not have (column "fitted", SSEE / ΛCDM). $\Delta\chi^2 = \chi^2_{\mathrm{SSEE}} - \chi^2_{\Lambda\mathrm{CDM}}$. "$N$" is the number of degrees of freedom where the log records it. [a]Seven nuisance parameters minimised identically in both models. [b]TATT intrinsic alignments (the survey's baseline); with NLA the difference is $\Delta\chi^2 = 0.41$.

Probe	Type	N	fitted	χ^2_{SSEE}	$\chi^2_{\Lambda\mathrm{CDM}}$	$\Delta\chi^2$
A. Background fixed in both models						
Pantheon+	SN	1579	0 / 0	1386.62	1387.33	−0.70
DES-SN5YR	SN	1819	0 / 0	1629.44	1632.41	−2.97
Union3	SN	21	0 / 0	23.85	26.37	−2.52
Union3.1	SN	21	0 / 0	26.60	28.83	−2.24
ACT DR6	CMB lensing	10	0 / 0	14.01	14.20	−0.20
SPT-3G D1 KK[a]	CMB lensing	10	0 / 0	17.76	18.31	−0.54
DES Y3[b]	3×2pt	–	0 / 0	516.11	517.29	−1.18
DESI DR2 BAO	BAO	13	0 / 0	11.41	31.98	−20.58
B. ΛCDM fits parameters that SSEE does not have						
DESI DR2 BAO	BAO	13 / 11	0 / 2	11.41	10.54	0.87
Planck plik_lite	CMB	–	2 / 6	1003.59	1003.60	−0.01
Planck plik (full)	CMB	–	2 / 6	2768.45	2771.23	−2.78
KiDS-Legacy	shear	–	0 / 1	417.97	418.95	−0.98

4 The CMB and the effective field theory

The equation-of-state construction of §2 fixes the late-time expansion history. To test the framework against the cosmic microwave background and large-scale structure we must specify its behaviour at the level of linear perturbations. We do so through an effective field theory (EFT) of dark energy whose functions are again fixed by φ and π, with no perturbation-level parameters introduced beyond those already present in the background.

4.1 The effective field theory of the dark sector

In the EFT-of-dark-energy language [Cheung et al., 2008, Gubitosi et al., 2013, Bellini and Sawicki, 2014] the linear dynamics of a single scalar degree of freedom are controlled by four functions α_{K} (kineticity), α_{B} (braiding), α_{M} (Planck-mass run) and α_{T} (tensor speed excess). For the operative Lagrangian of Paper 7, $K(X) = c_1 X + c_2 X^2$, minimally coupled and with no potential, they are

$$\alpha_{\mathrm{T}} = \alpha_{\mathrm{M}} = \alpha_{\mathrm{B}} = 0, \qquad \alpha_{\mathrm{K}} = 15.591336, \tag{9}$$

so that gravitational waves propagate at the speed of light ($\alpha_{\mathrm{T}} = 0$, consistent with GW170817 [Abbott et al., 2017]) and the only non-trivial function is the kineticity, evaluated with the dark-energy *density* $\Omega_{\mathrm{DE}} = 1 - \Omega_m = 0.691119$.

A distinct quantity, often confused with it, enters the Hubble screening of Paper 9:

$$s_K \equiv -\frac{1}{\rho_\phi}\frac{d|p_\phi|}{dN}\bigg|_{w=w_0} = -3w_0(1+w_0) = 3\, s_{\mathrm{DE}}\, s_m = 0.403302, \tag{10}$$

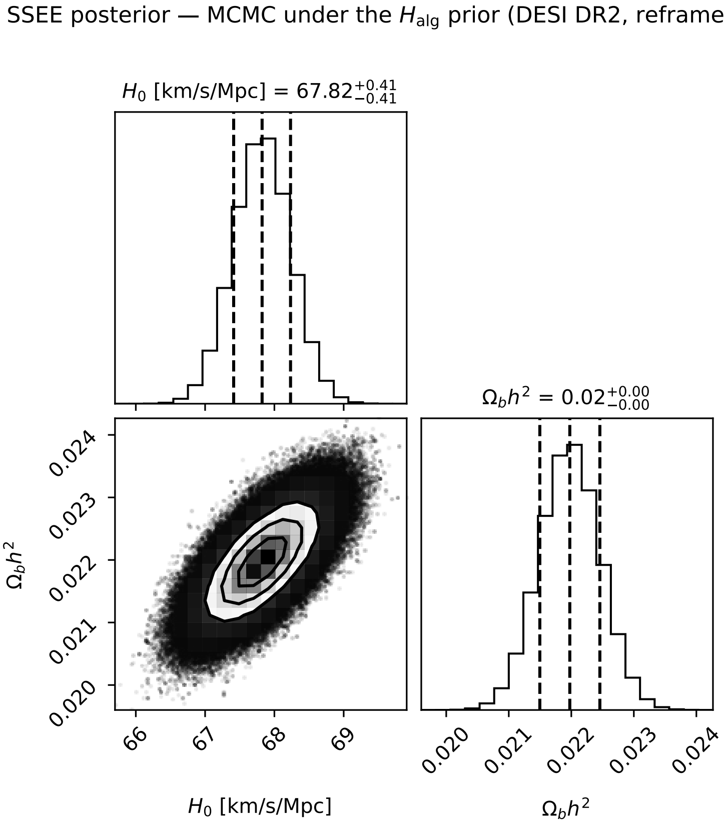

Figure 3: Marginalised posterior from the MCMC analysis (100 walkers × 25,000 steps). With the entire dark-energy sector fixed by φ and π, the chain samples only H_0 and $\Omega_b h^2$ against DESI DR2 BAO with $H_0^{\rm glob}$ as prior. The posterior is $H_0 = 67.82 \pm 0.41$ $\rm km\,s^{-1}\,Mpc^{-1}$ and $\Omega_b h^2 = 0.02198 \pm 0.00048$ (Eq. 8), agreeing with the anchor to 0.34σ. Dashed lines mark the prior centres ($H_0^{\rm glob} = 67.962$ $\rm km\,s^{-1}\,Mpc^{-1}$ and the BBN $\Omega_b h^2 = 0.02218$ Schöneberg [2024]); the 16/50/84% quantiles are indicated.

the fractional dilution rate of the dark-energy *pressure* per e-fold, evaluated with w_0 held fixed: the term from the evolution of $w(a)$ is not part of s_K, which is a number of the algebra rather than the full rate (Paper 9). Here $s_{\rm DE} = |w_0| = 0.839950$ and $s_m = 1 + w_0 = 0.160050$ are saturations — numbers of the equation of state, carrying no H — not densities. Because s_K is pure equation of state, the screening amplitude $f_{\rm scr} = s_K/(3\,\rm MIRA)$ is invariant under changes of the expansion anchor, whereas a kineticity would not be. The Lagrangian above contains no coupling of the scalar to matter. What remains open is tracked as OP-24: the same action fixes w_0 but does not produce $w_a = -0.669975$.

4.2 Inflationary sector: α-attractor

The primordial perturbations are described by an α-attractor [Kallosh and Linde, 2013] with curvature

$$\alpha = \frac{\varphi^4}{3} = 2.284701. \tag{11}$$

For the attractor universality class, $n_s = 1 - 2/N_*$ and $r = 12\alpha/N_*^2$. We take the number of e-folds to be $N_* = 2\varphi^7 = 58.068884$. The coefficient 2 is not a free choice. Within the observationally allowed window $[50, 60]$ the products $m\varphi^n$ with small integer m are $2\varphi^7$, $3\varphi^6$, $5\varphi^5$ and $8\varphi^4$; of these, $m = 2$ is the only one for which the spectral index comes out as a pure power of φ, precisely because that 2 cancels against the 2 in $n_s = 1 - 2/N_*$ (the others give $1 - \varphi^{-6.84}$, $1 - \varphi^{-6.90}$ and

Spectrum	SSEE χ^2_r	ΛCDM χ^2_r	N
TT	1.042	1.043	1971
TE	1.040	1.040	1967
EE	1.040	1.039	1967
$\phi\phi$ (lensing)	0.720	0.757	9

Table 5: CMB confrontation against Planck 2018. SSEE reproduces the acoustic spectra at a quality indistinguishable from ΛCDM, with acoustic peaks at $\ell = 220, 536, 813$. The combined ΔBIC(SSEE $-$ ΛCDM) $= -35.0$ over $N = 5914$ data points (diagonal, ω_m-direct) reflects the two-parameter economy of the background.

$1 - \varphi^{-6.88}$). The same coefficient simultaneously renders r a pure power. $N_* = 2\varphi^7$ is therefore the unique member of that family for which *both* observables reduce to pure powers of φ, and the two resulting identities are exact:

$$n_s = 1 - \varphi^{-7} = 0.965558, \qquad r = \varphi^{-10} = 8.130619 \times 10^{-3}. \tag{12}$$

The spectral index sits squarely on the Planck 2018 measurement ($n_s = 0.9649 \pm 0.0042$; Planck Collaboration, 2020a); the tensor-to-scalar ratio $r = \varphi^{-10}$ is a pre-committed, falsifiable prediction (§6) well below the current bound $r < 0.036$ [BICEP/Keck Collaboration, 2021] and within reach of LiteBIRD [LiteBIRD Collaboration, 2023] and CMB-S4. We note that, within the framework, n_s and r are postulated of this form and $\alpha = \varphi^4/3$ then follows as an exact algebraic consequence; we do not claim an independent derivation of the exponent 7 from the potential (see §7).

4.3 Confrontation with Planck 2018

With the background of §2 and the EFT functions of Eq. (9), we computed the CMB angular power spectra with `CAMB` [Lewis et al., 2000] (the Paper 3 pipeline) and compared against Planck 2018 [Planck Collaboration, 2020b] (TT, TE, EE and lensing); the peak positions are cross-checked with `CLASS` [Lesgourgues, 2011]. Table 5 reports the reduced χ^2.

Which likelihood the model comparison rests on. We are explicit about this because the two numbers differ and only one of them is a model comparison. The χ^2_r of Table 5 and the associated ΔBIC $= -35.0$ use a *diagonal* covariance. That is adequate for displaying spectral agreement, but not for selecting between models: neglecting the Planck bandpower correlations overstates the number of independent data points, and since the BIC penalty enters as $k \ln N$, it inflates precisely the term that rewards the smaller k. We therefore also evaluate the compressed Planck likelihood `plik_lite` (TT+TE+EE, 613 bandpowers, nuisance parameters marginalised by the Planck Collaboration) together with the low-ℓ TT and EE likelihoods, for $N = 669$ data points, which gives $\chi^2_{\text{SSEE}} = 1003.586$ against $\chi^2_{\Lambda\text{CDM}} = 1003.596$ (with its own $\sum m_\nu = 0.06$ eV) with $\{A_s, \tau\}$ minimized in both—an equal fit ($\Delta\chi^2 = -0.010$, statistically zero) for four fewer free parameters—and

$$\Delta\text{BIC}(\text{SSEE} - \Lambda\text{CDM}) = -26.03 \qquad (\texttt{plik\_lite}, k = 2 \text{ vs } 6). \tag{13}$$

The diagonal -35.0 is retained for continuity with Paper 3 and should be read as illustrative. One residual dependence would remain—`plik_lite` marginalises its nuisance parameters within a ΛCDM analysis—so we also evaluated the *full* `plik` likelihood (TTTEEE + lowl TT + lowl EE + lensing, $N = 2354$) with the $\sim$20 nuisance parameters free, and take the *minimum* χ^2 of each model:

$$\chi^2_{\text{SSEE}} = 2768.4, \quad \chi^2_{\Lambda\text{CDM}} = 2771.2, \quad \Delta\text{BIC} = -33.83 \qquad (\texttt{plik full}, k = 2 \text{ vs } 6). \tag{14}$$

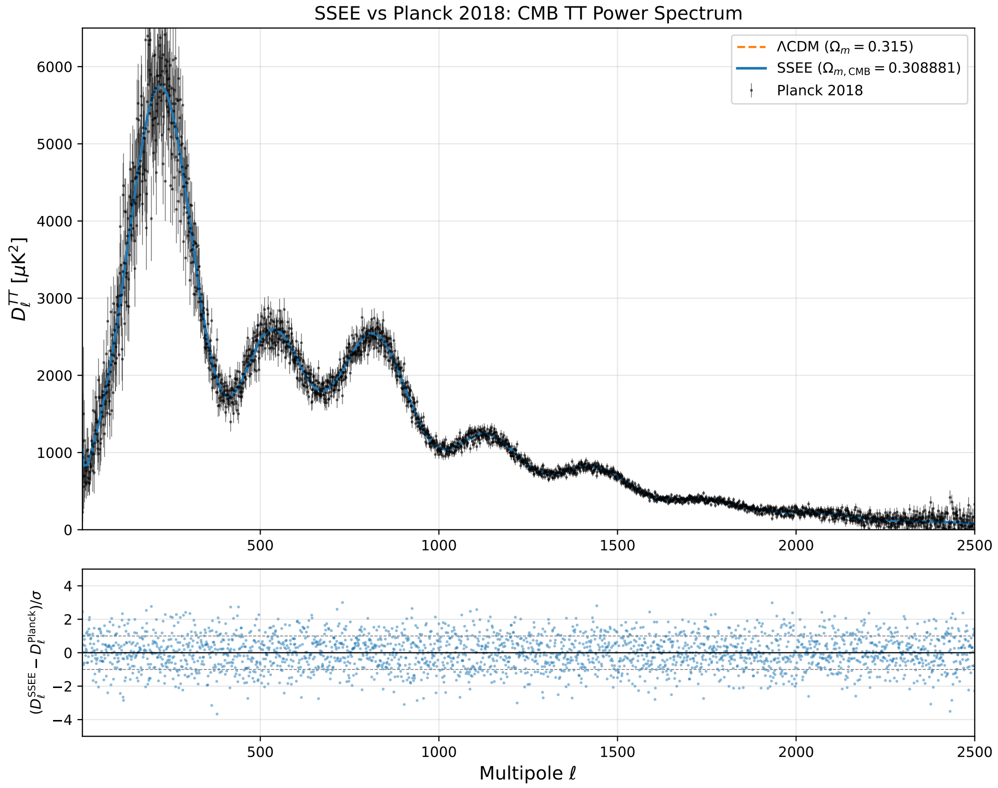

Figure 4: CMB temperature angular power spectrum computed with `CAMB` (the Paper 3 pipeline) for the SSEE background and EFT functions (Eq. 9), compared against Planck 2018. With the dark-energy sector fixed by φ and π, SSEE reproduces the acoustic peaks at $\ell = 220$, 536, 813 at a quality indistinguishable from ΛCDM (Table 5).

The two fits are indistinguishable ($\Delta\chi^2 = -2.78$); the gain is the parameter count. Counting conservatively $k = 4$ for SSEE gives -18.31. As a validation, a separate chain that also samples H_0 returns $H_0 = 67.88 \pm 0.10$ km s^{-1} Mpc^{-1}, 0.81σ from $H_0^{\mathrm{glob}} = 67.962$: fixing H_0 costs no fit quality.

The CMB with the single matter density. Running the Boltzmann solver (CLASS) at the ω_m-direct matter density $\Omega_{\mathrm{m}}^{\mathrm{cosm}} = 0.308881$ places the first three acoustic peaks at $\ell = 220$, 536, 813 (ΛCDM: 220, 536, 813), an RMS deviation of 0.18% from ΛCDM; CAMB returns the same peaks (Paper 3). The density is not fitted to the CMB: it follows from the ω_m-direct identity ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, fixed by φ, π; §7). An earlier version contrasted this with a run that put the equation-of-state number $1 + w_0 = 0.160050$ in the density slot; $1 + w_0$ is not a density, so that run is not a version of the model and the comparison is withdrawn.

5 Structure growth: no tension on raw data

The same linear EFT predicts the growth of structure, and here we report the situation honestly, including a correction to what earlier versions of this document reported. With the Poisson source evaluated at the CMB-calibrated density $\Omega_{\mathrm{m}}^{\mathrm{cosm}} = 0.308881$ (ω_m-direct, the model's single matter density), the growth factor is essentially that of ΛCDM, $G \equiv D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032$, and the single-sector amplitude ceiling (CLASS forward) is

$$\sigma_8 = 0.8149, \qquad S_8 = 0.827 \tag{15}$$

when the primordial amplitude A_s is held at the Planck-preferred value. That configuration was previously reported as a 3.5σ tension with KiDS-1000 [Asgari et al., 2021, DES Collaboration, 2022], and a two-sector φ-DM extension was proposed to resolve it. **Both the tension and the extension are withdrawn here**, for two independent reasons.

First, fixing A_s to Planck is not a neutral choice: A_s is one of the two free parameters this model retains at CMB level, so a number obtained that way reports the Planck–shear disagreement rather than a property of the model. That disagreement is real and is not resolved here — asking this model's own CMB fit to set A_s, rather than borrowing Planck's, still leaves 3.5σ against the shear determination. Second, the target itself was model-conditioned: the published S_8 is a compressed statistic whose data-reduction pipeline assumes a ΛCDM background, so fitting to it compares a non-ΛCDM model against a partly-ΛCDM number.

Measured instead against the 225 *raw* KiDS-1000 $\xi_\pm$ data points, with the background fixed entirely by the algebra, a *single* unpartitioned matter sector $\Omega_m = 0.308881$, and only A_s and the eight official nuisance parameters free, a converged MCMC ($R - 1 = 0.019$, 42711 effective samples) gives

$$\sigma_8 = 0.7449 \pm 0.0186, \qquad S_8 = 0.7559 \pm 0.0189, \tag{16}$$

i.e. 0.10σ from the KiDS-1000 value 0.759 ± 0.024 — and, in the like-for-like comparison, $\Delta S_8 = 0.0012$ from the ΛCDM control run through the same pipeline (0.7571 ± 0.0194), with $\chi^2_{\text{min}} = 265.44$ over 216 degrees of freedom. **There is no S_8 tension to resolve, and therefore nothing for a second sector to do.**

The same test against KiDS-Legacy. The measurement above uses KiDS-1000. Repeating it against the KiDS-Legacy release—357 raw $\xi_\pm$ points, Wright et al. (2025) (arXiv:2503.19441), published sixteen months before the present analysis, over which we claim no temporal priority—closes the sector. With the same algebraic background and A_s free, the shear asks for $\log(10^{10}A_s) = 3.0255 \pm 0.0396$, against the 3.04483 that the CMB fixes *under that same background*: a distance of 0.49σ. The control with the Planck background held equally rigid gives 1.01σ. Fixing A_s to the CMB value therefore costs nothing:

$$\chi^2 = 417.97 \;\; \text{(8 free, all nuisance, } \textit{none} \text{ cosmological)} \quad \text{vs.} \quad 418.34 \;\; (A_s \text{ free}), \tag{17}$$

over the same 357 points, giving $\Delta\text{BIC} = +6.24$ in favour of the fixed-amplitude model, and $+6.86$ against the ΛCDM-background control (418.95). With no free cosmological parameter left, $S_8 = 0.8273$ is a prediction rather than a fit, against 0.8265 ± 0.0176 measured—0.04σ.

Two cautions belong with this result. The three χ^2 values span 1.0 over 357 points: **the shear does not discriminate between these models**, and we do not claim that it does. What separates them is the number of parameters each had to spend and whether A_s stays consistent between the CMB and the growth epoch. And the earlier KiDS-1000 release asked for $\log(10^{10}A_s) = 2.864 \pm 0.050$, 3.6σ from the same target; the two releases differ from each other by 2.5σ. What changed is the data—the $n(z)$ calibration, by the collaboration's own Appendix I—not the model. A future release returning to the lower amplitude would reopen the sector.

The φ-DM component and its particle are retracted independently of this: the subtraction that defined the second sector's density, $\Omega_{\varphi\text{DM}} = \Omega_{m,\text{CMB}} - s_m$, took a measured density minus $1 + w_0$ — a number from the equation of state, not a density. The subtraction is arithmetically well formed and physically empty, so the particle had nothing to be made of.

The honest statement. What resolves S_8 here is not additional freedom — the background is *more* constrained in this model than in ΛCDM, with A_s the only quantity able to move — but the removal of two things that should not have been there: a fixed A_s, and a target statistic carrying ΛCDM in its reduction. The cost of the correction is real and worth stating plainly: a pre-registered, genuinely falsifiable $P(k)$ prediction is gone, withdrawn not because data refuted it but because the construction generating it was not physical. What is gained is one postulate fewer

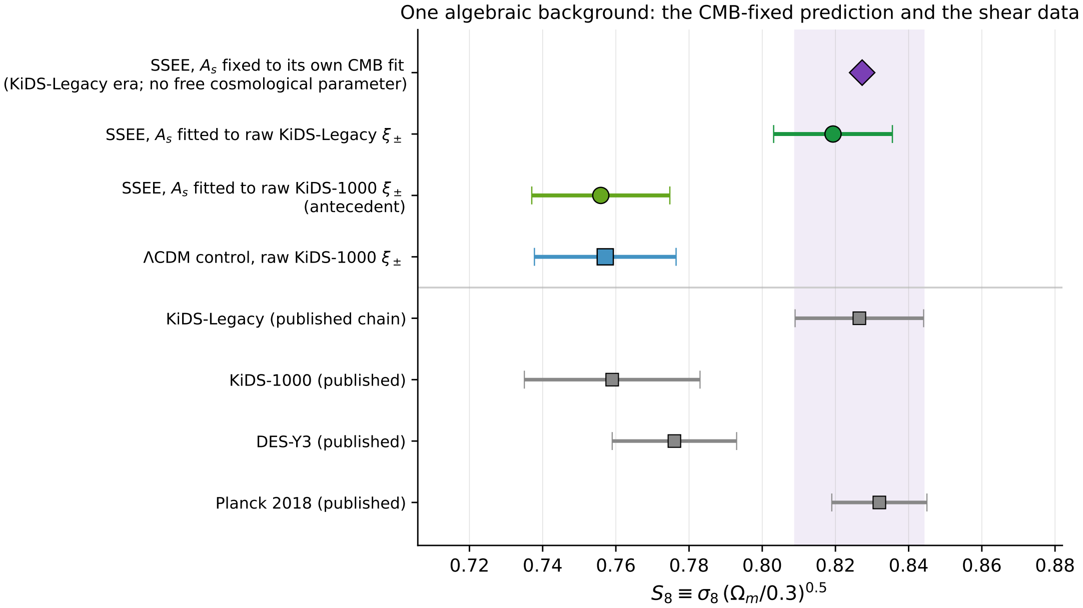

Figure 5: S_8 on the algebraic background of SSEE, identical in every SSEE row; only the dataset asked to fix A_s changes. Purple diamond: the prediction with A_s fixed to the model's own CMB fit — no free cosmological parameter — $S_8 = 0.8273$, 0.04σ from the KiDS-Legacy posterior (0.8265 ± 0.0176, purple band). Green: A_s fitted to the raw KiDS-Legacy $\xi_\pm$ (0.8193 ± 0.0162) and, as antecedent, to the raw KiDS-1000 $\xi_\pm$ (0.7559 ± 0.0189, 0.10σ from KiDS-1000). Blue: the ΛCDM control on the same raw KiDS-1000 data (0.7571 ± 0.0194). Grey: published KiDS-Legacy, KiDS-1000, DES-Y3 and Planck 2018 constraints with 1σ errors. The shift between the two KiDS releases is in the data, not in the model.

and four open problems closed by dissolution. A full non-linear treatment still requires N-body simulations beyond the scope of this work.

6 Falsifiability and the prediction register

The defining property of a parameter-frugal framework is that, once the algebraic invariants are fixed, the numbers it predicts are not free to move. A fit can always be improved by adding a parameter; SSEE has none to add. We therefore commit, in advance, to a register of quantitative predictions, each attached to the observation that will confirm or refute it. We separate the register into the *core* predictions that follow directly from the dimensionless invariants φ and π, and the growth, UV and strong-gravity results that build on them; no extension sector is involved in either (the one proposed earlier is retracted, §7).

6.1 Master register: one line per observable

Before the two tiers in detail, Table 6 gives the entire suite on a single page: every confronted observable, the sector it belongs to, the paper that derives it, its algebraic value, and its epistemic status (postdiction / retrodiction / prediction / measured input). The horizontal rule separates the *core* tier—fixed by φ and π alone—from the results that build on it (growth, UV completion, strong gravity). This is the navigation map for the whole programme; each row is expanded in the paper named in its last-but-one column and in the Predictive Register of Paper 1.

Observable	Sector	Value	Paper	Status
Core tier — clean from φ, π (no fitted coefficient)				
$w_0 = -T_r/M_v$	Background (DE)	-0.839950	P1,2,4	Postdiction
$w_a = -P_{\rm sc}/I_g$	Background (DE)	-0.669975	P1,2,4	Postdiction
(w_0, w_a) vs DESI DR3	Background (DE)	same fixed point	P1,2	**Prediction (pre-reg.)**
$H_0^{\rm glob} = H_0^{\rm SH0ES}(1 - f_{\rm scr})$	Background (DE)	$67.962\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (target $3(\varphi + \pi)^2$)	P9,10	Derived (SH0ES input)
BAO distances, background fixed	Background	$\chi^2 = 11.41/13$ (ΛCDM-Planck 31.98; Tab. 4)	P2	Postdiction
$\omega_c = {\rm KAL}_0\, \omega_b\, n_s$	CMB	0.119514	P1,3	Retrodiction
$\Omega_{\rm m}^{\rm cosm} = \omega_m/h^2$	CMB	0.308881	P1,3	Retrodiction
$n_s = 1 - \varphi^{-7}$	Inflation / CMB	0.965558	P1,3	Retrodiction
$r = \varphi^{-10}$	Inflation	8.1×10^{-3}	P1	**Prediction**
$\alpha_T = \alpha_M = \alpha_B$	EFT	0 (exact)	P7	Retrodiction
$\alpha_K = 3\Omega_{\rm DE}(5 - 3w_0)$	EFT	15.591336	P7	Derived (unobservable on linear scales)
$\gamma_{\rm IS}$ (Israel–Stewart)	Growth	0.5504	P5	Retrodiction
$f\sigma_8$ (BOSS DR12, raw multipoles)	Growth	$\Delta\chi^2 = -2.03$ vs. ΛCDM, equal freedom	P6	Retrodiction
Built on the core — growth, UV completion, strong gravity (no extension sector)				
S_8 (KiDS-Legacy, A_s fixed by CMB)	growth	0.8273 predicted vs. 0.8265 ± 0.0176 (0.04σ)	P6	**Prediction**
$M = (5\varphi^8 \rho_{\rm crit})^{1/4}$	UV compl.	9.68 meV	P10	Conditional (Conj. C.1)
Lensing (canonical EFT)	Strong gravity	GR+DM (uncond.)	P8	Retrodiction
Measured input (not derived)				
$A_s,\ \tau$	CMB amplitude	sampled ($k = 2$)	P1,3	Measured
$H_0^{\rm SH0ES}$ (abs. scale)	Background	single measured input	P1,9	Measured

Table 6: Suite-wide master register: one row per observable across the ten-paper programme. "Postdiction": algebraic result fixed by the closed dictionary, then compared to already-public data (no claim of temporal priority). "Retrodiction": derived independently, then compared to an existing measurement. "Prediction": fixed now, tested against not-yet-released data. "Measured": an input, not a derivation. "Internal check": a same-paper consistency quantity, not itself tested against external data. No optimisation over the algebraic formula space was performed. The detailed per-sector registers follow below and in the Predictive Register of Paper 1.

6.2 Core predictions (clean from φ, π)

These quantities carry no fitted coefficient. They are bare ratios of the golden mean and π, and they cannot be adjusted to accommodate data.

Observable	SSEE value	Algebraic form	Falsifying probe
w_0	-0.839950	$-T_r/M_v$	DESI Y3 / Euclid BAO+SNe
w_a	-0.669975	$-P_{\rm sc}/I_g$	DESI Y3 / Euclid BAO+SNe
n_s	0.965558	$1 - \varphi^{-7}$	Planck / CMB-S4
r	8.1306×10^{-3}	φ^{-10}	LiteBIRD / CMB-S4
c_s^2	0.021284	$(M_v - T_r)/(5M_v + 3T_r)$	DESI / Euclid growth
$\alpha_T, \alpha_M, \alpha_B$	0	exactly zero	GW standard sirens, ISW

Table 7: Core prediction register. Each value is fixed by φ and π alone, with no free parameter available to absorb a discrepancy. The tensor ratio $r = \varphi^{-10}$ sits an order of magnitude below the current bound $r < 0.036$ and is the cleanest single falsification target: a detection above $\sim 10^{-2}$, or a robust upper bound below $\sim 8 \times 10^{-3}$, refutes the inflationary sector.

The tensor-to-scalar ratio deserves emphasis. Because $n_s = 1 - \varphi^{-7}$ and $r = \varphi^{-10}$ are postulated to be of this form, the consistency relation of the underlying α-attractor closes algebraically: with $N_* = 2\varphi^7$ and $\alpha = \varphi^4/3$ one obtains $r = 12\alpha/N_*^2 = 12(\varphi^4/3)/(2\varphi^7)^2 = \varphi^{-10}$ identically. The prediction is thus internally locked rather than independently adjustable. Its value, $r \simeq 8.1 \times 10^{-3}$, lies squarely within the reach of LiteBIRD and CMB-S4, which target $\sigma(r) \sim 10^{-3}$. SSEE will be decisively tested by the next generation of B-mode experiments.

A pre-registered DESI DR3 test. The equation-of-state point is fixed forever at $(-0.840, -0.670)$; each DESI release is therefore a fresh blind test of the *same* target. Against DR2 it sits at 0.24σ (Pantheon+; §2.2), inside the 68% region, which a fitted model can always arrange but a rigid one need not. We pre-commit, in advance of the DR3 release (expected

2027), to the following quantitative expectations for DESI DR3, whose uncertainties are forecast to improve by $\sim 1.5\times$: if the DR2 central values persist, the point will sit at $\sim 0.5\sigma$ of the DR3+CMB+Pantheon+-class posterior (still inside 1σ), while the Union3/DESY5-class combinations would sharpen to ~ 2.2–2.7σ —DR3 will therefore *discriminate among the supernova compilations* whose spread already dominates the range quoted in §2.2. The falsification line is pre-registered *now*, before the DR3 release, and fixed to a specific supernova compilation so that no post-hoc choice of sample can rescue the model: a joint exclusion of $(-0.840, -0.670)$ at $>3\sigma$ in the DR3 w_0–w_a plane for the $\mathrm{DR3} + \mathrm{CMB} + \mathit{Pantheon+}$ combination —the same Pantheon+ compilation used for the headline agreement throughout this work—refutes the algebraic background. We will additionally report the Union3- and DESY5-class combinations for completeness, but the pre-committed falsifier is the Pantheon+ line named here. No parameter exists with which to absorb such an outcome.

The vanishing of the EFT slip functions $\alpha_T = \alpha_M = \alpha_B = 0$ is equally sharp: SSEE predicts a gravitational-wave speed identical to light ($c_T = c$) to all orders in the dark sector, no running of the effective Planck mass, and no braiding. Any measured departure—an anomalous standard-siren distance, a time-varying G_{eff}, or a non-zero braiding signature in the ISW–galaxy cross-correlation—falsifies the canonical EFT presented in §4.

7 Honest accounting and outstanding challenges

A framework is only as credible as its honesty about what it does not yet explain. We therefore state plainly the inputs the model still requires, the exact parameter count, and the principal open problems—not as caveats buried in a footnote, but as the research agenda that the present work both motivates and constrains.

7.1 What is measured, not derived: the absolute scale

SSEE does not predict the absolute energy density of the dark sector. The only dimensionful input is a measured one, the local expansion rate H_0^{SH0ES} (*Postulate D*, as in Paper 1); everything with units descends from it through the screening cascade. What the algebra fixes is the *shape* of the dark-energy sector—its equation-of-state ratios w_0, w_a, the matter density ω_m, and the EFT function profile—not its amplitude. We make no claim to have solved the cosmological-constant problem; we have moved its free numbers from six tuned cosmological parameters down to a single measured normalisation plus the dimensionless invariants φ and π. This is a reduction in arbitrary inputs, not their elimination.

7.2 Parameter count: $k = 2$ versus 6

We claim a *minimal-parameter* framework, never a zero-parameter one. The honest accounting is the following. ΛCDM carries six fitted parameters $(\Omega_b h^2, \Omega_c h^2, \theta_*, \tau, n_s, A_s)$. SSEE fixes four of them algebraically from φ and π: the baryon density ω_b (OP-1), the CDM density via the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, the spectral index $n_s = 1 - \varphi^{-7}$, and the expansion anchor $H_0 = 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (whose value in these units coincides with $3(\varphi+\pi)^2$; derived, Paper 9). This leaves exactly two free CMB parameters—the primordial amplitude A_s and the optical depth τ—a $k = 2$ versus $k = 6$ comparison, which is the count used in the Paper 3 Bayesian model selection. The dark-energy ratios (w_0, w_a) and the EFT functions are predicted, not fitted. On the dark-sector extension there is nothing left to promise: the φ-DM mass coefficient (OP-9) and the non-minimal coupling ξ (OP-11) are retracted with the sector itself, and both close by dissolution, tracked separately as open problems rather than counted among the fit parameters.

7.3 Open problems as bridges to future work

The open problems that bear on this work are listed below; one of them has closed since earlier versions and is kept for the record. They define the path from the present minimal-parameter framework toward a fully derived theory.

- **The CMB matter density's algebraic origin (OP-1).** With the former MIRA matter bridge *dissolved* (the ω_m-direct reframe, §2.4; OP-8 retired), the CMB matter fraction $\Omega_{m,\mathrm{CMB}} = 0.308881$ now rests directly on two algebraic inputs: the baryon density $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ and the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514$ (0.40σ from Planck). With it the CLASS run of §4.3 reproduces the acoustic peaks of ΛCDM. What remains open is a first-principles derivation of these algebraic forms from the field action; the burden has moved from "derive MIRA" to "ground ω_b and the ω_c identity." The values are not fitted, but their UV origin is open.
- **The φ-dark-matter mass (OP-9): closed by dissolution.** This entry previously recorded an algebraic mass m_φ whose UV origin was open. The particle itself is now retracted, so there is no multiplier left to derive and the open problem closes without having been solved. The free-streaming step in $P(k)$ that descended from it is withdrawn as a prediction. OP-10, OP-11 and OP-12 close the same way.

The growth-of-structure status of §5 no longer mirrors this structure. The S_8 "tension" at 0.827 was an artefact of holding A_s at the Planck value while comparing against a ΛCDM-conditioned summary statistic; fitted to raw shear data with a single sector it is 0.7559 ± 0.0189, matching the ΛCDM control on the same data (0.7571 ± 0.0194) with four fewer free parameters. Against the later KiDS-Legacy release the growth sector needs no free amplitude either: the A_s the shear asks for sits 0.49σ from the one the CMB fixes under the same background, and fixing it predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured. No extension sector is required, and the one proposed is retracted. The non-linear treatment required to settle the matter definitively still demands N-body simulations beyond this work's scope.

None of these gaps touch the algebraic core of Table 7: the dark-energy equation of state, the spectral index, the tensor ratio, and the vanishing EFT slip functions stand on the dimensionless invariants alone and are falsifiable today.

8 Conclusions

We have presented a minimal-parameter framework in which the shape of the dark sector is fixed by the two dimensionless constants φ and π. The late-time equation of state emerges as bare ratios of structural invariants, $w_0 = -0.839950$ and $w_a = -0.669975$, and matches the DESI DR2 posterior at 0.24σ (Pantheon+) without a single fitted dark-sector parameter. The effective-field-theory description is canonical, with vanishing tensor, running, and braiding functions, $\alpha_K = 15.591336$ and $c_s^2 = 0.021284$; the inflationary sector predicts $n_s = 0.965558$ and $r = 8.1 \times 10^{-3}$; and a Boltzmann calculation with its single matter density places the acoustic peaks where ΛCDM does.

The framework's strength is its frugality and its consequent falsifiability. Its core predictions—the dark-energy ratios, the spectral index, the tensor ratio, and the vanishing slip functions—rest on φ and π alone and will be tested decisively by DESI Y3, Euclid, LiteBIRD, and CMB-S4 within this decade. We have been equally explicit about its limits: the absolute scale is a measured input (the local expansion rate) and the algebraic origin of the CMB matter density (ω_b, ω_c) is unresolved. The growth amplitude, once reported here as a tension with weak lensing, is not one on raw shear data: with A_s fixed to the model's own CMB fit, the growth sector predicts $S_8 = 0.8273$ against KiDS-Legacy's 0.8265 ± 0.0176. These define a concrete research agenda rather than fatal flaws, and none of them touch the algebraic core.

Whether the deep agreement between a handful of golden-ratio combinations and the measured dark sector reflects an underlying structural principle or a remarkable set of coincidences is precisely the question that the prediction register of §6 is designed to settle. The next generation of cosmological surveys will provide the answer.

A The two lineage laws of the constant dictionary

The look-elsewhere test of §2.3 rests on the dictionary being a *closed, finite* set fixed in advance, rather than an open search space. This appendix states the generative rules that make it so. We present them as a descriptive grammar of the Genesis set [Almeida Vallejo, 2026]: they reproduce the dictionary exactly, but their *dynamical* origin is not claimed and remains the open mechanism of §7. Stating them explicitly is what turns "55 named invariants" from an assertion into a checkable construction.

Seeds and scaffold. Everything descends from two dimensionless seeds, $\varphi = (1+\sqrt{5})/2$ (UV) and π (IR). Their sum is the scaffold $\Omega = \varphi + \pi = 4.759627$, and their mean is the bifurcation point $\beta \equiv \mathrm{BIAL} = (\varphi+\pi)/2 = 2.379813$ at which the dictionary splits into two branches.

Law I — the copy law (φ-branch). The φ-branch is generated by a single root, the screening amplitude $\mathrm{AURA} = \beta + \varphi = 3.997847$, which propagates *only* by integer multiples and the dyadic half. These are the dimensional ceilings:

$$\begin{gathered} \mathrm{MIRA} = \tfrac{1}{2}\mathrm{AURA}, \quad \mathrm{AURA}, \quad \mathrm{DUAL} = 2\,\mathrm{AURA}, \\ \mathrm{TRIAL} = 3\,\mathrm{AURA} = T_r, \quad \mathrm{CUARTAL} = 4\,\mathrm{AURA}. \end{gathered} \tag{18}$$

The denominator of w_0 is $M_v = \varphi + \pi + K_v$, a structural convergence built by lineage on K_v (KRYSTOS $= \varphi + \pi + \Omega$); it *coincides in value* with 3Ω but is not the scaffold tripled—the construction, not the magnitude, is what fixes its identity. With the copy-law wall $T_r = \mathrm{TRIAL} = 3\,\mathrm{AURA}$ as numerator,

$$|w_0| = \frac{T_r}{M_v} = \frac{\mathrm{TRIAL}}{\varphi + \pi + \mathrm{KRYSTOS}} = 0.839950, \tag{19}$$

which reduces *numerically* to AURA/Ω—an identity of value between distinct named entities, not a construction by scaffold multiples. So w_0 and w_a are ratios internal to the closed dictionary, not two independent draws.

Law II — the no-self-sum law (π-branch). The π-branch is generated additively: each child is a sum of previously defined parents, subject to the constraint that *a node never adds to itself.* Thus $\mathrm{KAL} = \beta + \pi$, P_{sc} (PYROS) $= \Omega + \varphi$, $\mathrm{IGNIS} = \pi + P_{\mathrm{sc}}$, and so on, building up to the convergences. A corollary is that two distinct lineages may land on the *same value* without being the same entity: $\mathrm{IGNIS} = \pi + \mathrm{PYROS}$ (disruptive, π-branch) and K_v (KRYSTOS) $= \varphi + \pi + \Omega$ (structural) both equal 9.519253, yet are distinct nodes; likewise $\mathrm{SOLAR} = \mathrm{MAR} = 7.90122$. The look-elsewhere count of §2.3 treats these honestly, deduplicating by value so the 55 named nodes collapse to 25 distinct numbers. The second equation-of-state ratio lives entirely on this branch,

$$|w_a| = \frac{P_{\mathrm{sc}}}{\mathrm{IGNIS}} = \frac{\mathrm{PYROS}}{\pi + \mathrm{PYROS}} = 0.669975, \tag{20}$$

where the denominator is the π-branch node IGNIS, not the structural K_v, though the two coincide in value.

Closure. The set is finite because both laws terminate at the four-dimensional integration invariant $M_v = \varphi + \pi + K_v = 14.278880$ (coinciding in value with 3Ω), the maximal dimensional invariant; no node exceeds it. This termination—not a choice made to fit data—is what bounds the dictionary to a fixed, enumerable set and licenses the conservative look-elsewhere denominator of §2.3. Any future extension would proceed only by the same copy law (the planned dimensional copies QUINTAL–DECAL), which, as noted in Table 2, add no accidental hit within the observational band of either w_0 or w_a.

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Data and Code Availability

The SSEE framework is fully transparent and reproducible. The complete code repository, including the CLASS and CAMB configuration files, the COBAYA MCMC chains, the explicit record of Open Theoretical Problems (OP), and the cosmological validation ledgers, are publicly available at the GitHub repository: `https://github.com/mikealmeida1721/SSEE-Cosmologia` and permanently archived on Zenodo, DOI 10.5281/zenodo.19654118 (concept DOI, all versions).

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